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General Motors

McCombs School of Business

University of Texas at Austin

Week 02
Capstone Team 8

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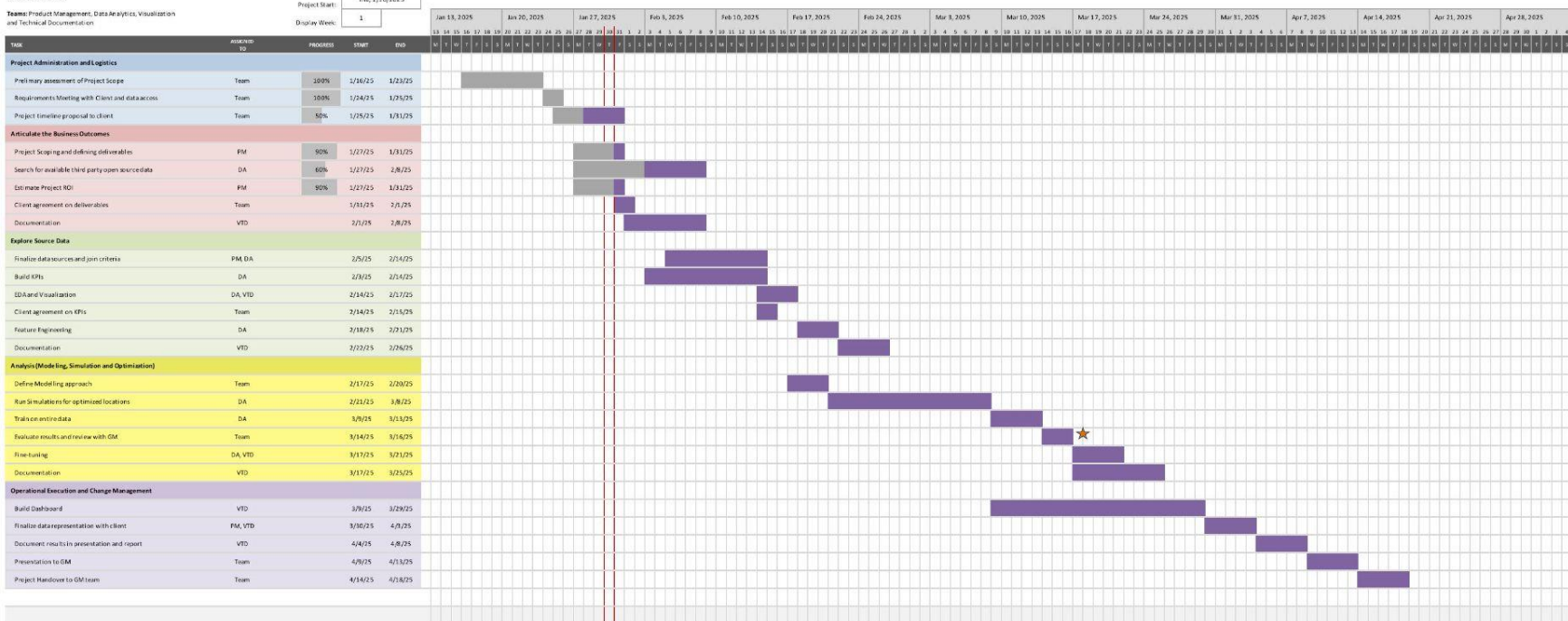
Executive Summary

1. Project Plan showcasing 13 weeks of progression with broad sections, milestones and team assignments
2. The state of Tennessee was chosen for its strong EV adoption potential and cost-effectiveness.
3. ROI for the project was drafted showcasing a positive NPV of \$0.3M and payback period of 3.8 years
4. Review of target data and KPIs to finalize business outcomes
5. Five different open source datasets were explored and the team saw usability from three of them
6. GM provided dataset was evaluated and follow up questions and requests for latest numbers
7. Feedback

1. Project Plan

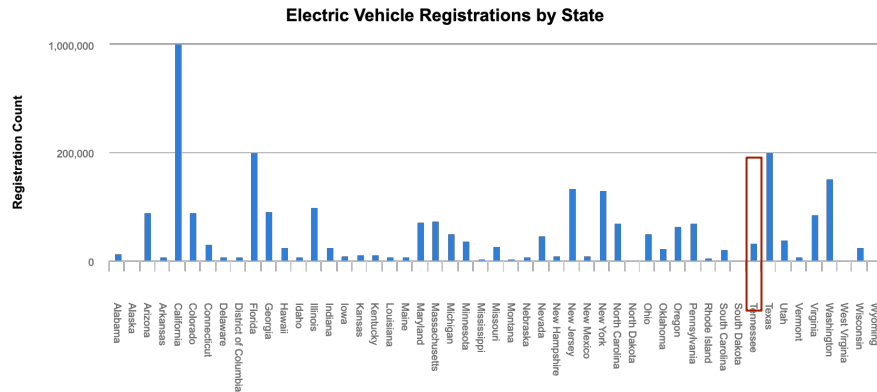
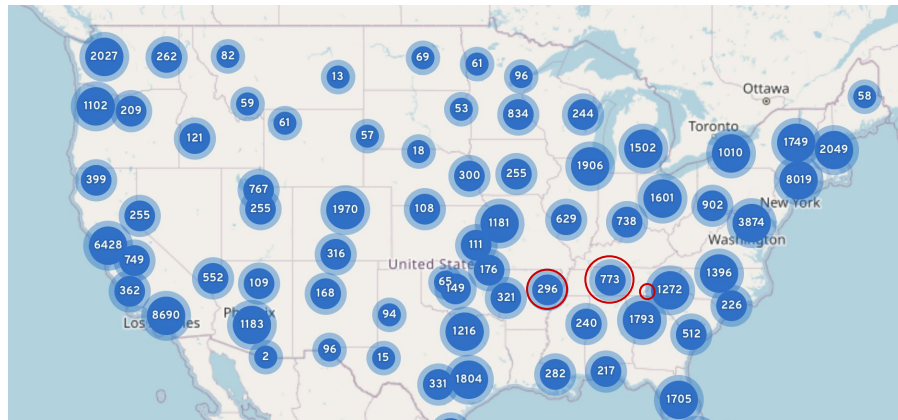
Powering the future through smart EV expansion

Capstone Team 8: Muhammad Ibrahim, Sarah Lee, John Hwang, Rohan Giri, Varsha Manju Jayakumar, Ammar Mustafa
 General Motors



2. State: Tennessee

- GM's **largest facility** is located in Spring Hills, TN which is also responsible for producing Ultium batteries for EV vehicles
- Tennessee ranks **above the median** in terms of EV adoption rates. (29.88%)
- 3 large cities (Nashville, Memphis, Knoxville) with overall population of 7.2 million (22% under 18 years)
- One of the few states that are allocated **federal grants** for EV charging stations (\$88M) under the National Electric Vehicle Infrastructure (NEVI) program.



3. ROI Calculation

Overall Costs :

- Development Costs: \$250,000
- Installation Costs: \$4,000,000 (100 chargers, \$40,000 each)
- Annual Maintenance Costs: \$350,000

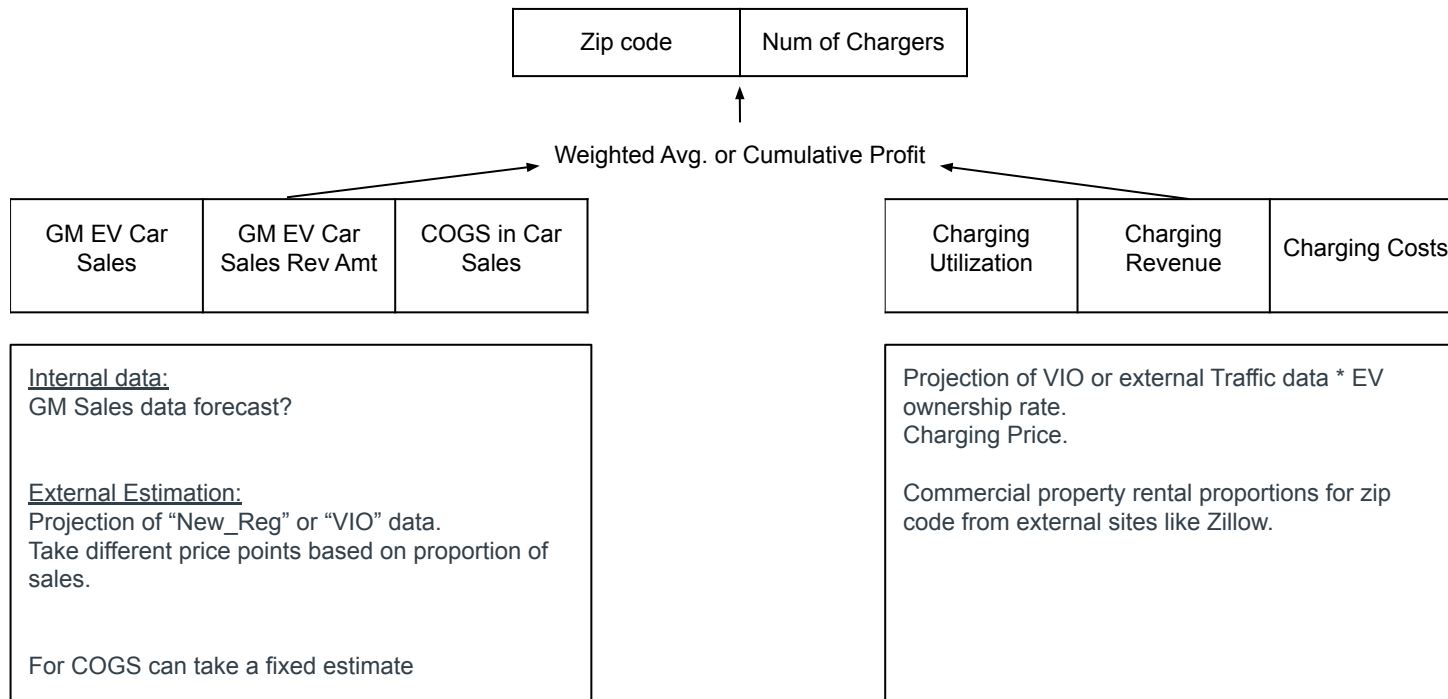
Benefits:

- Projected Increase in EV Sales: \$1.2M annually (30% growth)
- Charger Utilization Revenue: \$500K annually (30% growth)
- Environmental Impact: 2,000 tons of CO2 reduced annually (\$200K value)

Financial Metrics:

- ROI (1-year): 27%
- NPV (3-year): \$368,613.82
- Payback Period: ~3.87 years

4. Target data and KPIs



5. Open Source Data sets

Our team evaluated 5 different open source data sets:

1. EV Infrastructure data
2. Tennessee EV registration data
3. Traffic data
4. Charging stations data
5. Tesla supercharger data

5.1 Existing EV Infrastructure

- This dataset contains information about electric vehicle (EV) charging stations across Tennessee.
- Why?
 - Identify gaps and avoid oversaturation
- **Station Name & Location:** Includes the station name, street address, city, and ZIP code for identifying charging points.
- **Charging Infrastructure:** EV Level 1, Level 2, and DC Fast Charger Count: Specifies the number of chargers available at different power levels.
- **EV Network:** Identifies the charging network (e.g., ChargePoint, Tesla, Electrify America).
- **Access Days & Time:** Indicates whether the station is available to the public and operational hours.
- **Restricted Access:** Specifies if the station is limited to certain users (e.g., workplace-only chargers).
- **EV Pricing:** Contains information on whether the charging station is free or paid.
- **Source:** Alternative Fuels Data Center

5.2 TN EV Registration

- This dataset contains records of electric vehicle (EV) registrations across Tennessee
- Why?
 - Helps prioritize areas for potential adoption
- **Registration Date & County:** Helps track EV adoption trends over time and by geographic region.
- **Vehicle Make & Model:** Identifies the specific EVs registered
- **Drivetrain Type:** Differentiates between Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).
- **Vehicle Gross Vehicle Weight Rating (GVWR) Category:** Classifies vehicles into Light-Duty (Class 1-2A) and others, indicating their size and usage.
- **Source:** Atlas EV hub

5.3 Traffic data

- This dataset contains records of traffic across Tennessee
- Why?
 - Helps identify strategic charger placement
- **Annual Average Daily Traffic (AADT):** Measures the average number of vehicles that travel on a specific road segment daily.
- **Vehicle Miles Traveled (VMT):** Estimates the total distance traveled by vehicles on a road segment.
- **Design Hourly Volume Percent:** Represents the percentage of traffic occurring during peak hours.
- **Vehicle Classification:** Breakdown of traffic by vehicle type, including passenger pickups, single-unit trucks, and multiple-unit trucks.
- **Geospatial & Log Mile Data:** Includes beginning and ending log miles to specify road segments, as well as shape length data for mapping.
- **Source:** TN state government.

6. GM Data - Follow up

- **Demographic data** → Given 2020 American Community Survey data, updated demographic data required
- **Fuel_Elec_Ratio column** → static value of 29.88
- **VIO & New_Reg** → Is Catalyst a third party company or data system of GM?
- **VIO** → From 1990 till when (2022)? need confirmation on the availability of recent data
- **New_Reg (2022)** → need confirmation on the availability of recent data
- **Intender_Projected_2022_Total_BEV_Rated 4_0+** → static value of 58
- **Intender_Projected_2022_Total_BEV_PCT_Top Box** → blank values
- **Charging station** → Limited and sparse data; majority fields are blank. Updated data is critical for meaningful insights
- **Solar system** → Source: <http://web.stanford.edu/group/deepsolar/home.html> (covers data only up to 2018). Updated data required.
- **Search features** → Placeholder columns currently contain blank values

Feedback